



Manual Scene Setup Guide

This guide walks through setting up the game scenes entirely in the Unity Editor (without the auto-bootstrap script).

Prerequisites

1. Open the project in Unity 2021.3+
 2. Set up custom tags (Edit → Project Settings → Tags and Layers):
 - `Player` , `Enemy` , `PlayerBullet` , `EnemyBullet` , `PowerUp`
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Part 1: Create Prefabs

1.1 Generate Sprites First

- Go to **Tools** → **Generate Placeholder Sprites** (requires `SpriteGenerator.cs`)
- Or use your own sprite assets imported as **Sprite (2D and UI)**, Pixels Per Unit = 32

1.2 Player Bullet Prefab

1. **GameObject** → **2D Object** → **Sprite** → name it `PlayerBullet`
2. Assign `BulletPlayer` sprite
3. Set Tag to `PlayerBullet`
4. Add Components:
 - `BoxCollider2D` → Is Trigger: ☒, Size: (0.15, 0.3)
 - `Rigidbody2D` → Gravity Scale: 0, Body Type: Kinematic
 - `BulletController` script
5. **Drag from Hierarchy into** `Assets/Prefabs/` to create prefab
6. Delete from scene

1.3 Enemy Bullet Prefab

1. Same as above but use `BulletEnemy` sprite
2. Tag: `EnemyBullet`
3. Save as `Assets/Prefabs/EnemyBullet.prefab`

1.4 Enemy Prefabs (create 4)

For each enemy type:

Prefab Name	Sprite	Tag
BasicEnemy	EnemyBasic	Enemy
FastEnemy	EnemyFast	Enemy
TankEnemy	EnemyTank	Enemy
ShooterEnemy	EnemyShooter	Enemy

For each:

1. **GameObject** → **2D Object** → **Sprite**
2. Assign the correct sprite
3. Set Tag to `Enemy`
4. Add Components:
 - `BoxCollider2D` → Is Trigger: ☒, Size: (0.8, 0.8)
 - `Rigidbody2D` → Gravity Scale: 0, Body Type: Kinematic
 - `EnemyController` script
5. Drag to `Assets/Prefabs/`
6. Delete from scene

1.5 Power-Up Prefabs (create 3)

Prefab Name	Sprite	Type (in Inspector)
PowerUpWeapon	PowerUpWeapon	Weapon Upgrade
PowerUpShield	PowerUpShield	Shield
PowerUpHealth	PowerUpHealth	Health

For each:

1. **GameObject** → **2D Object** → **Sprite**
2. Set Tag to `PowerUp`
3. Add Components:
 - `CircleCollider2D` → Is Trigger: ☒, Radius: 0.35
 - `Rigidbody2D` → Gravity Scale: 0, Body Type: Kinematic
 - `PowerUpController` script → set **Type** dropdown
4. Drag to `Assets/Prefabs/`
5. Delete from scene

1.6 Player Prefab

1. **GameObject** → **2D Object** → **Sprite** → name `Player`
2. Assign `PlayerShip` sprite
3. Set Tag to `Player`
4. Position: (0, -3.5, 0)
5. Add Components:
 - `BoxCollider2D` → Is Trigger: ☒, Size: (0.8, 0.8)

- RigidBody2D → Gravity Scale: 0, Body Type: Kinematic
 - PlayerController script
6. Create child empty object named FirePoint at local position (0, 0.6, 0)
 7. In PlayerController Inspector:
 - Drag PlayerBullet prefab → **Bullet Prefab**
 - Drag FirePoint → **Fire Point**
 - Drag the SpriteRenderer → **Sprite Renderer**
 8. Optionally save as prefab in Assets/Prefabs/
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Part 2: Set Up GameScene

2.1 Camera Setup

1. Select **Main Camera**
2. Set Projection: **Orthographic**
3. Orthographic Size: **5.5**
4. Background Color: Dark blue/black (0.01, 0.01, 0.06)

2.2 Background

1. Create two large sprites using the Background image
2. Position them at Y=0 and Y=12 to tile
3. Set Sorting Order to -100
4. Add ParallaxBackground script to each:
 - Scroll Speed: 1.0 (and 1.2 for second)
 - Reset Y: -12
 - Start Y: 12

2.3 Stars (Optional)

Create 20-30 small sprites with the Star sprite:

- Random positions across the screen
- Random alpha (0.3-1.0)
- Random scale (0.5-1.5)
- Add ParallaxBackground with varying scroll speeds (0.5-3.0)

2.4 Player

- Place the Player in the scene at (0, -3.5, 0)
- Ensure all Inspector references are wired

2.5 Game Manager

1. **GameObject** → **Create Empty** → name GameManager
2. Add GameManager script
3. Wire references:
 - **Enemy Spawner**: (create next)
 - **UI Manager**: (create next)
 - **Player**: drag Player from hierarchy

2.6 Enemy Spawner

1. **GameObject** → **Create Empty** → name EnemySpawner

2. Add `EnemySpawner` script
3. Wire prefab references in Inspector:
 - **Basic Enemy Prefab** → `BasicEnemy` prefab
 - **Fast Enemy Prefab** → `FastEnemy` prefab
 - **Tank Enemy Prefab** → `TankEnemy` prefab
 - **Shooter Enemy Prefab** → `ShooterEnemy` prefab
 - **Enemy Bullet Prefab** → `EnemyBullet` prefab
 - **Power Up Prefabs** → array of 3 power-up prefabs

2.7 Audio Manager

1. **GameObject** → **Create Empty** → name `AudioManager`
 2. Add `AudioManager` script
 3. (Optional) Add AudioClips to Sound Effects array
 4. (Optional) Assign Background Music clip
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Part 3: Set Up UI Canvas

3.1 Create Canvas

1. **GameObject** → **UI** → **Canvas**
2. Canvas: Render Mode = **Screen Space - Overlay**
3. Canvas Scaler: UI Scale Mode = **Scale With Screen Size**, Reference = 1920×1080
4. Add `UIManager` script to Canvas

3.2 HUD Panel

1. Create child **Panel** named `HUDPanel`
2. Set Image color to transparent (alpha = 0)
3. Add children:
 - **Text** `ScoreText` : anchored top-left, "Score: 0", size 28
 - **Text** `WaveText` : anchored top-right, "Wave: 0", size 28
 - **Text** `HealthText` : anchored bottom-left, "HP: 5/5", size 24
 - **Slider** `HealthSlider` : anchored bottom-left, min=0, max=5, value=5
 - **Text** `WaveAnnouncement` : centered, size 48, yellow, initially inactive

3.3 Pause Menu Panel

1. Create child **Panel** named `PauseMenuPanel`
2. Set Image color to semi-transparent black (0, 0, 0, 0.7)
3. Add `PauseMenuController` script
4. Add children:
 - **Text**: "PAUSED", size 48, centered at top
 - **Button** `ResumeBtn` : "Resume"
 - **Button** `RestartBtn` : "Restart"
 - **Button** `MenuBtn` : "Main Menu"
 - **Button** `QuitBtn` : "Quit"
5. Wire button references in `PauseMenuController` Inspector
6. **Set panel inactive** (uncheck checkbox at top of Inspector)

3.4 Game Over Panel

1. Create child **Panel** named `GameOverPanel`
2. Set Image color to semi-transparent black (0, 0, 0, 0.8)
3. Add `GameOverController` script
4. Add children:
 - **Text** `GameOverTitle` : "GAME OVER", size 56, red
 - **Text** `FinalScore` : "Final Score: 0", size 32
 - **Text** `HighScore` : "High Score: 0", size 28
 - **Button** `RestartBtn` : "Play Again"
 - **Button** `MenuBtn` : "Main Menu"
 - **Button** `QuitBtn` : "Quit"
5. Wire references in `GameOverController` Inspector
6. **Set panel inactive**

3.5 Wire UIManager References

Select the Canvas and in the UIManager Inspector, drag:

- `HUDPanel` , `ScoreText` , `WaveText` , `HealthText` , `HealthSlider`
- `PauseMenuPanel`
- `GameOverPanel` , `FinalScore` , `HighScore`
- `WaveAnnouncement`

3.6 Event System

If not auto-created: **GameObject** → **UI** → **Event System**

Part 4: Create MainMenuScene

1. **File** → **New Scene** → save as `Assets/Scenes/MainMenuScene.unity`
2. Set camera background to dark blue
3. Create **Canvas** with:
 - **Text** `TitleText` : "SPACE SHOOTER", size 64, white, centered near top
 - **Text** `HighScoreText` : "High Score: 0", size 28, centered
 - **Button** `StartButton` : "START GAME", centered
 - **Button** `QuitButton` : "QUIT", centered below start
4. Add `MainMenuController` to Canvas
5. Wire references: `TitleText`, `HighScoreText`, `StartButton`, `QuitButton`

Part 5: Build Settings

1. **File** → **Build Settings**
2. Add both scenes:
 - `MainMenuScene` at index 0
 - `GameScene` at index 1
3. Platform: **PC, Mac & Linux Standalone**
4. Target: **Windows**, Architecture: **x86_64**
5. Click **Build** and choose output folder

Hierarchy Summary (GameScene)

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GameScene
├── Main Camera
├── Background_0 [ParallaxBackground]
├── Background_1 [ParallaxBackground]
├── Star_0..Star_29 [ParallaxBackground]
├── Player [PlayerController, SpriteRenderer, BoxCollider2D, Rigid-
body2D]
├── FirePoint
├── GameManager [GameManager]
├── EnemySpawner [EnemySpawner]
├── AudioManager [AudioManager]
├── GameCanvas [Canvas, CanvasScaler, GraphicRaycaster, UIManager]
├── HUDPanel
│   ├── ScoreText
│   ├── WaveText
│   ├── HealthText
│   ├── HealthSlider
│   └── WaveAnnouncement
├── PauseMenuPanel [PauseMenuController] (inactive)
│   ├── PauseTitle
│   ├── ResumeBtn
│   ├── RestartBtn
│   ├── MenuBtn
│   └── QuitBtn
├── GameOverPanel [GameOverController] (inactive)
│   ├── GameOverTitle
│   ├── FinalScore
│   ├── HighScore
│   ├── RestartBtn
│   ├── MenuBtn
│   └── QuitBtn
└── EventSystem
  
```